Sector decomposition for an anisotropic oscillatory integral

1 Setting and overview

1.1 The integral

Let $\ell_1, \dots, \ell_d \geq 0$ and, for $s > 0$, let $A_s = \operatorname{diag}(s^{\ell_1}, \dots, s^{\ell_d})$. For $t \geq 1$ and $\xi \in \mathbb{R}^d$ we define the phase

$$\psi(t, \xi) = \int_1^t |A_s \xi| \, ds.$$

We study the oscillatory integral

$$I(t, x) = \int_{\mathbb{R}^d} e^{i(x \cdot \xi - \psi(t,\xi))} \, |A_t \xi|^{-1} \, \chi(\xi) \, d\xi,$$

where χ is a smooth cutoff supported in the annulus $\{\frac{1}{2} \leq |\xi| \leq 2\}$. We want to bound $I(t, x)$ for large t, uniformly in x.

1.2 The difficulty and the plan

The matrix A_t stretches each frequency coordinate by a different power of t. So there is no single large parameter in front of the phase. We handle this in two moves.

First we split frequency space into d smooth sectors, according to which weighted coordinate $t^{\ell_j}|\xi_j|$ is the largest. On sector k we use the leading coordinate ξ_k and the $d - 1$ ratios $\eta_j = t^{\ell_j} \xi_j / (t^{\ell_k} \xi_k)$ as new variables. In these variables the sector integral becomes an outer integral in ξ_k of an inner oscillatory integral in η with a single large parameter $\lambda = t^{\ell_k + 1} |\xi_k|$ (Section 2).

Second we bound the inner integral by stationary phase. The phase is strictly concave in η, and its curvature in each direction is at least of order 1; in borderline directions it reaches order $\log t$ when the leading coordinate takes over early enough (Section 3). Its higher derivatives are not bounded uniformly in t, but they are once we weight them correctly (Section 4). The stationary phase theorem needs its constants to be bounded uniformly in t. At unit scale in η they are not; at the natural scale of each direction, $(\lambda L_j)^{-1/2}$ with $L_j = 1$ or $L_j = \log t$, they are. Section 5 checks this, and Section 6 assembles the main result, Theorem 5. The outer integral in ξ_k is not treated here.

1.3 Conventions

In all estimates $t \geq 2$. The index k is fixed, $n = d - 1$, and i, j, m run over $\{1, \dots, d\} \setminus \{k\}$. Multi-indices $\alpha = (\alpha_j)_{j \neq k}$ have n entries, and δ_i is the multi-index with a 1 in position i and 0 elsewhere. We write C, c for positive constants that depend only on d and $\ell_1, \dots, \ell_d$ and may change from line to line. C_α may also depend on α.

2 Reduction to one sector

2.1 Smooth sectors

Roughly, sector k is where $t^{\ell_k}|\xi_k|$ is the largest of the weights $t^{\ell_1}|\xi_1|, \dots, t^{\ell_d}|\xi_d|$. Since $|A_t\xi|^2$ is the sum of the squared weights, the largest weight is at least $|A_t\xi|/\sqrt{d}$. The stationary phase theorem needs a smooth amplitude, so we use a smooth version of this splitting. Fix smooth $\rho_1, \dots, \rho_d$ on $\mathbb{S}^{d-1}$ with $\sum_k \rho_k = 1$ and $\rho_k(p) = 0$ unless $|p_k| \geq \frac{1}{2\sqrt{d}}$. Such a partition exists since $\max_k |p_k| \geq 1/\sqrt{d}$ on $\mathbb{S}^{d-1}$. Then $I = \sum_k I_k$, where I_k is I with the extra factor $\rho_k(A_t\xi/|A_t\xi|)$ in the integrand. On the support of this factor

$$t^{\ell_k}|\xi_k| \;\geq\; \frac{|A_t\xi|}{2\sqrt{d}}, \tag{1}$$

so weight k is comparable to the largest one. We fix k.

2.2 New variables

For $j \neq k$ we introduce the ratios

$$\eta_j = \frac{t^{\ell_j}\xi_j}{t^{\ell_k}\xi_k}, \qquad \text{that is} \qquad \xi_j = t^{\ell_k - \ell_j}\xi_k\,\eta_j,$$

which compare the weight of coordinate j with that of coordinate k. We write $\eta = (\eta_j)_{j\neq k} \in \mathbb{R}^n$ and use (ξ_k, η) as coordinates.

2.3 The integrand in the new variables

We rewrite each factor of the integrand in terms of ξ_k and η.

The factor $|A_t\xi|^{-1}$. Since $A_t\xi = t^{\ell_k}\xi_k\big(e_k + \sum_{j\neq k}\eta_j e_j\big)$,

$$|A_t\xi| = t^{\ell_k}|\xi_k|\,(1+|\eta|^2)^{1/2}, \qquad |A_t\xi|^{-1} = t^{-\ell_k}|\xi_k|^{-1}(1+|\eta|^2)^{-1/2}.$$

The phase ψ. In the same way, for $0 < s \leq t$,

$$|A_s\xi| = s^{\ell_k}|\xi_k|\Big(1 + \sum_{j\neq k}\frac{s^{2\ell_j}\xi_j^2}{s^{2\ell_k}\xi_k^2}\Big)^{1/2}.$$

We measure time in units of t: substitute $s = \sigma t$, so $ds = t\,d\sigma$ and $s^{2(\ell_j-\ell_k)}\xi_j^2/\xi_k^2 = \sigma^{2(\ell_j-\ell_k)}\eta_j^2$. This gives

$$\psi(t,\xi) = t^{\ell_k+1}|\xi_k|\,G(t,\eta), \qquad G(t,\eta) = \int_{1/t}^1 \sigma^{\ell_k}\Big(1 + \sum_{j\neq k}\sigma^{2(\ell_j-\ell_k)}\eta_j^2\Big)^{1/2} d\sigma.$$

So the phase is a large factor $t^{\ell_k+1}|\xi_k|$ times a function G of η, whose only dependence on t is through the lower limit $1/t$.

The linear term. We aim at a bound uniform in x, so we may replace x by $A_t x$ without changing $\sup_x |I(t,x)|$. The linear part of the phase becomes

$$A_t x\cdot\xi = x_k t^{\ell_k}\xi_k + \sum_{j\neq k} t^{\ell_j}x_j\xi_j = x_k t^{\ell_k}\xi_k + t^{\ell_k}\xi_k\sum_{j\neq k}x_j\eta_j.$$

The measure. For fixed ξ_k,

$$d\xi = |\xi_k|^n \prod_{j\neq k} t^{\ell_k-\ell_j}\; d\xi_k\, d\eta = t^{\,d\ell_k-\sum_{j=1}^d \ell_j}\,|\xi_k|^n\, d\xi_k\, d\eta.$$

2.4 The sector integral

Collecting the formulas above,

$$I_k = t^{\,d\ell_k-\sum_j \ell_j-\ell_k}\int_{\mathbb{R}} e^{ix_k t^{\ell_k}\xi_k}\,|\xi_k|^{d-2}\,J(\xi_k)\,d\xi_k, \qquad J(\xi_k)=\int_{\mathbb{R}^n} e^{i\lambda\Phi(\eta)}\,b(\xi_k,\eta)\,d\eta,$$

with the large parameter and the phase

$$\lambda = t^{\ell_k+1}|\xi_k|, \qquad \Phi(\eta)=\frac{\xi_k}{|\xi_k|}\,\frac{1}{t}\sum_{j\neq k} x_j\eta_j - G(t,\eta),$$

and the amplitude

$$b(\xi_k,\eta)=(1+|\eta|^2)^{-1/2}\,\tilde\chi(\xi_k,\eta)\,\rho_k\Big(\frac{\xi_k}{|\xi_k|}\,\frac{e_k+\sum_{j\neq k}\eta_j e_j}{(1+|\eta|^2)^{1/2}}\Big), \qquad \tilde\chi(\xi_k,\eta)=\chi\Big(\xi_k e_k+\sum_{j\neq k} t^{\ell_k-\ell_j}\xi_k\eta_j\,e_j\Big).$$

The rest of these notes bounds the inner integral J.

The phase. Φ depends on x and on the sign of ξ_k only through its linear part. So

$$\partial_\eta^\alpha\Phi = -\partial_\eta^\alpha G \qquad (|\alpha|\geq 2), \tag{2}$$

and every bound on derivatives of order at least two is uniform in x.

The support of the amplitude. On the support of $b(\xi_k,\cdot)$ three facts hold. The kth component of the argument of ρ_k is $\pm(1+|\eta|^2)^{-1/2}$, so $|\eta|\leq 2\sqrt{d}$. The cutoff $\tilde\chi$ forces $|\xi_m|\leq 2$ for every m, that is, $|\xi_k|\leq 2$ and

$$|\eta_m| \leq \frac{2\,t^{\ell_m-\ell_k}}{|\xi_k|} \qquad (m\neq k). \tag{3}$$

Finally, (1) and $|\xi|\geq\frac12$ give $t^{\ell_k}|\xi_k|\geq t^{\ell_{\min}}/(4\sqrt{d})$ with $\ell_{\min}=\min_j\ell_j$. Hence

$$\frac{t^{\ell_{\min}-\ell_k}}{4\sqrt{d}} \;\leq\; |\xi_k| \;\leq\; 2, \qquad \lambda \;\geq\; \frac{t^{1+\ell_{\min}}}{4\sqrt{d}}.$$

So λ is large. Note, however, that a derivative ∂_{η_j} falling on $\tilde\chi$ produces a factor $t^{\ell_k-\ell_j}\xi_k$, which is large when $\ell_j<\ell_k$. This will be the main obstacle in Section 5.

2.5 Types of directions

How each direction $j\neq k$ is treated depends on the number

$$\kappa_j = 2\ell_j-\ell_k+1.$$

We call direction j

- *good* if $\ell_j \geq \ell_k$, and *bad* if $\ell_j < \ell_k$;
- *oscillatory* if $\kappa_j > 0$, *borderline* if $\kappa_j = 0$, and *flat* if $\kappa_j < 0$.

We write $\mathcal{O}$, $\mathcal{Z}$ and $\mathcal{F}$ for the sets of oscillatory, borderline and flat directions. Every good direction is oscillatory, since then $\kappa_j \geq \ell_k + 1 > 0$. So borderline and flat directions are bad. If there is a bad direction we put

$$\theta = \tfrac{1}{2} \min_{m \text{ bad}} (\ell_k - \ell_m) > 0.$$

In short: oscillatory directions gain from oscillation, flat directions gain from the small size of their support, and borderline directions gain from oscillation with an extra factor $(\log t)^{-1/2}$ when the curvature has time to build up.

3 The Hessian of the phase

Set

$$H(\sigma, \eta) = \Big(1 + \sum_{j \neq k} \sigma^{2(\ell_j - \ell_k)} \eta_j^2\Big)^{1/2}, \qquad \text{so that} \qquad G(t, \eta) = \int_{1/t}^{1} \sigma^{\ell_k} \, H(\sigma, \eta) \, d\sigma.$$

The goal of this section is to show that Φ is strictly concave in η, and to measure its curvature in each direction.

The quadratic form. Let $D(\sigma) = \operatorname{diag}\big(\sigma^{2(\ell_j - \ell_k)}\big)_{j \neq k}$. Since $\nabla_\eta H = H^{-1} D\eta$, differentiating twice and using (2) gives

$$-\nabla_\eta^2 \Phi = \nabla_\eta^2 G = \int_{1/t}^{1} \sigma^{\ell_k} H^{-3} \Big(H^2 D - (D\eta)(D\eta)^T \Big) \, d\sigma.$$

Fix $q \in \mathbb{R}^n$ and put $z = D^{1/2}\eta$, $u = D^{1/2} q$. Then $\langle Dq, q \rangle = |u|^2$, $\langle D\eta, q \rangle = \langle z, u \rangle$, and $|z|^2 = H^2 - 1$. Hence

$$-\langle \nabla_\eta^2 \Phi \, q, q \rangle = \int_{1/t}^{1} \sigma^{\ell_k} H^{-3} \Big(H^2 |u|^2 - \langle z, u \rangle^2 \Big) \, d\sigma.$$

By Cauchy–Schwarz, $\langle z, u \rangle^2 \leq |z|^2 |u|^2 = (H^2 - 1)|u|^2$, so the bracket lies between $|u|^2$ and $H^2 |u|^2$. Since $\sigma^{\ell_k} |u|^2 = \sum_j \sigma^{\kappa_j - 1} q_j^2$, this gives the two-sided bound

$$\sum_{j \neq k} q_j^2 \int_{1/t}^{1} \sigma^{\kappa_j - 1} H^{-3} \, d\sigma \;\leq\; -\langle \nabla_\eta^2 \Phi \, q, q \rangle \;\leq\; \sum_{j \neq k} q_j^2 \int_{1/t}^{1} \sigma^{\kappa_j - 1} H^{-1} \, d\sigma. \tag{4}$$

Strict concavity. For $q \neq 0$ the left side is positive. So $-\nabla_\eta^2 \Phi$ is positive definite and Φ is strictly concave in η, for every t, x and ξ_k.

Where the curvature comes from. Both sides of (4) are diagonal: the curvature in direction j is an integral of $\sigma^{\kappa_j - 1}$ against a power of H. On a dyadic time interval $[2^{-i-1}, 2^{-i}]$ where H stays bounded, this integral gains about $2^{-i\kappa_j}$. So the curvature of an oscillatory direction is built at late times, that of a borderline direction equally at every time scale, and that of a flat direction mostly at early times. We write

$$\Gamma_j(a) = \int_a^1 \sigma^{\kappa_j - 1} \, d\sigma \qquad (0 < a < 1).$$

Since $H \geq 1$, the upper bound in (4) gives $-\nabla_\eta^2 \Phi \leq \operatorname{diag}\big(\Gamma_j(1/t)\big)$, and

$$\Gamma_j(1/t) \leq 1/\kappa_j \quad (j \in \mathcal{O}), \qquad \Gamma_j(1/t) = \log t \quad (j \in \mathcal{Z}). \tag{5}$$

For a lower bound we need to know when H is bounded.

The switch time. For a bad direction m, the term $\sigma^{2(\ell_m - \ell_k)}\eta_m^2$ of H^2 is at most 1 exactly when $\sigma \geq |\eta_m|^{1/(\ell_k - \ell_m)}$. We call

$$\sigma_*(\eta) = \max_{m \text{ bad}} |\eta_m|^{1/(\ell_k - \ell_m)}$$

the switch time, with $\sigma_* = 0$ if there is no bad direction. In the original variables, $\sigma^{\ell_m - \ell_k}|\eta_m| = s^{\ell_m}|\xi_m|/(s^{\ell_k}|\xi_k|)$ with $s = \sigma t$. So $s_* = \sigma_* t$ is the time at which the weight $s^{\ell_k}|\xi_k|$ overtakes every bad weight $s^{\ell_m}|\xi_m|$.

Lemma 1 (Curvature)**.** *Let $R \geq 1$ and $|\eta| \leq R$, and put $\sigma_0 = \max\big(1/t, \min(\sigma_*(\eta), \frac{1}{2})\big)$. Then for all $q \in \mathbb{R}^n$,*

$$-\langle \nabla_\eta^2 \Phi(\eta)\, q, q\rangle \;\geq\; c_R \sum_{j \neq k} \Gamma_j(\sigma_0)\, q_j^2,$$

where $c_R > 0$ depends only on R, d and the ℓ_j. In particular:

(i) $\Gamma_j(\sigma_0) \geq c$ for every j, so $-\nabla_\eta^2 \Phi \geq c_R\, I$ on $|\eta| \leq R$;

(ii) if j is borderline and $\sigma_*(\eta) \leq t^{-1/2}$, then $\Gamma_j(\sigma_0) \geq \frac{1}{2}\log t$.

Proof. We show that $H \leq C_R$ on $[\sigma_0, 1]$. A good m contributes $\sigma^{2(\ell_m - \ell_k)}\eta_m^2 \leq R^2$. A bad m contributes at most 1 when $\sigma \geq \sigma_*$, and at most $4^{\ell_k - \ell_m} R^2$ when $\sigma \geq \frac{1}{2}$; one of the two holds on $[\sigma_0, 1]$. So $H^{-3} \geq c_R$ there, and restricting the lower bound in (4) to $[\sigma_0, 1]$ proves the inequality. For (i), $\sigma_0 \leq \frac{1}{2}$, so $\Gamma_j(\sigma_0) \geq \Gamma_j(\frac{1}{2}) > 0$. For (ii), $\kappa_j = 0$ gives $\Gamma_j(\sigma_0) = \log(1/\sigma_0)$, and $\sigma_0 \leq \max(1/t, t^{-1/2}) = t^{-1/2}$. □

Part (ii) is where $\log t$ enters. A borderline direction collects the same amount $\log 2$ of curvature from every dyadic time scale after the switch. If the switch comes before $t^{1/2}$ in physical time, about $\frac{1}{2}\log_2 t$ scales contribute. Together with (5), the curvature of a borderline direction is then exactly of order $\log t$. For flat directions Γ_j is even larger, but there the amplitude is too rough at the stationary phase scale, and this curvature is not used.

4 Higher derivatives of the phase

By (2) it suffices to bound $\partial_\eta^\alpha G$ for $|\alpha| \geq 2$. We first give a direct bound. Then we explain why it is not uniform in t, and we prove a weighted bound that is.

4.1 A direct bound

Reduction to one function. For fixed σ put $\mu_m = \sigma^{\ell_m - \ell_k}$, $y_m = \mu_m \eta_m$ and $\langle y\rangle = (1 + |y|^2)^{1/2}$. Then $H(\sigma, \eta) = \langle y\rangle$. Since each y_m is a multiple of η_m alone, the chain rule gives

$$\partial_\eta^\alpha H = \mu^\alpha \left(\partial_y^\alpha \langle y\rangle\right), \qquad \mu^\alpha = \prod_{m \neq k} \mu_m^{\alpha_m}.$$

Derivatives of $\langle y\rangle$. We claim that for $|\alpha| \geq 1$, $\partial_y^\alpha \langle y\rangle$ is a finite linear combination of terms

$$y^\beta \langle y\rangle^{1-2m} \qquad \text{with} \qquad m \geq 1, \quad 2m - |\beta| = |\alpha|.$$

For $|\alpha| = 1$ this holds, since $\partial_{y_i}\langle y\rangle = y_i \langle y\rangle^{-1}$. Differentiating $y^\beta \langle y\rangle^{1-2m}$ in y_i gives two terms: $\beta_i\, y^{\beta-\delta_i}\langle y\rangle^{1-2m}$, which vanishes if $\beta_i = 0$, and $(1-2m)\, y^{\beta+\delta_i}\langle y\rangle^{1-2(m+1)}$. In both terms $2m - |\beta|$ increases by one, which proves the claim. Since $|y^\beta| \leq \langle y\rangle^{|\beta|}$, each term is bounded by $\langle y\rangle^{|\beta|+1-2m} = \langle y\rangle^{1-|\alpha|}$. So $\langle y\rangle$ behaves like a function homogeneous of degree one, and

$$\left|\partial_\eta^\alpha H\right| \leq C_\alpha\, \mu^\alpha\, H^{1-|\alpha|}. \tag{6}$$

The bound. Differentiating under the integral sign and using $H \geq 1$,

$$\left|\partial_\eta^\alpha G(t,\eta)\right| \leq C_\alpha \int_{1/t}^1 \sigma^{\ell_k} \mu^\alpha \, d\sigma = C_\alpha \int_{1/t}^1 \sigma^{\,\ell_k + \sum_{m\neq k} \alpha_m(\ell_m - \ell_k)}\, d\sigma.$$

The last integral stays bounded as $t \to \infty$ only if the exponent of σ is larger than -1. When some $\ell_j < \ell_k$, this fails for large α_j.

4.2 Bad directions and weights

Why the direct bound fails. High derivatives in a bad direction j blow up at $\eta_j = 0$. Indeed, set all η_m with $m \neq j$ to zero and expand $H = (1 + \sigma^{2(\ell_j - \ell_k)}\eta_j^2)^{1/2}$ in powers of η_j. The coefficient of η_j^{2N} is $\binom{1/2}{N}\sigma^{2N(\ell_j-\ell_k)}$, so

$$\partial_{\eta_j}^{2N} G(t,0) = (2N)!\binom{1/2}{N}\int_{1/t}^1 \sigma^{\,\ell_k + 2N(\ell_j - \ell_k)}\, d\sigma.$$

Since $\ell_j < \ell_k$, the exponent is below -1 for N large, and the integral tends to infinity as $t \to \infty$. The reason is the switch of Section 3: near $\eta_j = 0$ the coordinate j dominates H at early times $\sigma \sim 1/t$, where $\sigma^{\ell_j - \ell_k}$ is huge. Away from $\eta_j = 0$ this does not happen.

The weights. We measure derivatives in units of the distance to these hyperplanes:

$$v_j(\eta) = 1 \;\; (j \text{ good}), \qquad v_j(\eta) = |\eta_j| \;\; (j \text{ bad}).$$

4.3 The weighted bound

By (6), $|\partial_\eta^\alpha H| \leq C_\alpha\, H \prod_{m\neq k}(\mu_m/H)^{\alpha_m}$. Two simple facts control each factor μ_m/H:

- if m is good, then $\mu_m \leq 1 \leq H$, so $\mu_m/H \leq 1$;
- if m is bad, then $H \geq \mu_m|\eta_m|$, so $\mu_m/H \leq |\eta_m|^{-1}$.

In both cases $\mu_m/H \leq v_m^{-1}$.

Lemma 2 (Weighted derivatives)**.** *Let $|\alpha| \geq 2$ with $\alpha_m = 0$ for $m \notin \mathcal{O}\cup\mathcal{Z}$, and choose $i, j \in \mathcal{O}\cup\mathcal{Z}$ (possibly $i = j$) such that $\alpha' = \alpha - \delta_i - \delta_j$ has nonnegative entries. Then*

$$\left|\partial_\eta^\alpha G(t,\eta)\right| \leq C_\alpha \left(\Gamma_i(1/t)\,\Gamma_j(1/t)\right)^{1/2} \prod_{m\neq k} v_m(\eta)^{-\alpha'_m}.$$

By (5), two derivatives in oscillatory directions are free, and a borderline one among them costs at most $(\log t)^{1/2}$. Each further derivative in a bad direction costs a factor $1/|\eta_m|$, and further derivatives in good directions cost nothing.

Proof. Since $|\alpha| = |\alpha'| + 2$, we can split the bound for $\partial_\eta^\alpha H$ as

$$\mu^\alpha H^{1-|\alpha|} = \frac{\mu_i \mu_j}{H} \prod_{m \neq k} \Big(\frac{\mu_m}{H}\Big)^{\alpha'_m} \leq \frac{\mu_i \mu_j}{H} \prod_{m \neq k} v_m^{-\alpha'_m}.$$

The product on the right does not depend on σ. It remains to integrate the first factor against $\sigma^{\ell_k} d\sigma$. Since $\sigma^{\ell_k} \mu_m^2 = \sigma^{\kappa_m - 1}$, the bound $H \geq 1$ and Cauchy–Schwarz give

$$\int_{1/t}^1 \sigma^{\ell_k} \, \frac{\mu_i \mu_j}{H} \, d\sigma \leq \Big(\int_{1/t}^1 \sigma^{\kappa_i - 1} d\sigma \Big)^{1/2} \Big(\int_{1/t}^1 \sigma^{\kappa_j - 1} d\sigma \Big)^{1/2} = \big(\Gamma_i(1/t)\, \Gamma_j(1/t) \big)^{1/2}. \qquad \square$$

5 The stationary phase theorem at the zoom scale

5.1 The theorem

For $b \in C_0^\infty(\mathbb{R}^n)$ with support K, let K_1 be the set of points at sup distance at most 1 from K, and set

$$M_m = \sum_{2 \leq |\alpha| \leq m} \sup_{K_1} |\partial^\alpha \Phi|, \qquad N_m = \sum_{|\alpha| \leq m} \sup_{K} |\partial^\alpha b|, \qquad a_0 = \inf_{K_1} \big| \det \nabla^2 \Phi \big|.$$

Theorem 3 (Alazard, Burq, Zuily, Theorem 3)**.** *Let Φ be real and smooth near K_1, with $a_0 > 0$ and $\nabla\Phi$ injective on K_1. Then for all $\lambda \geq 1$,*

$$\Big| \int_{\mathbb{R}^n} e^{i\lambda\Phi(\eta)} \, b(\eta) \, d\eta \Big| \leq \frac{C_n}{a_0} \big(1 + M_{n+2}\big)^{n/2} \, N_{n+1} \, \lambda^{-n/2},$$

where C_n depends only on n.

Two features of this bound matter for us. It involves no volume of K, so a large support costs nothing. And it sees Φ only through derivatives of order at least two, so the linear part of Φ, which carries x, plays no role.

5.2 Why the unit scale is not enough

Applied directly in η, Theorem 3 has its hypotheses satisfied for every fixed t: $a_0 \geq c$ by Lemma 1, and $\nabla\Phi$ is injective by concavity. But its constant is not uniform in t. Each ∂_{η_j} falling on $\tilde\chi$ produces the factor $t^{\ell_k - \ell_j} \xi_k$, so N_d can be as large as $t^{d(\ell_k - \ell_{\min})}$. By Section 4, the phase derivatives in M_{d+1} grow with t in bad directions. Borderline directions add a milder problem: Lemma 1 can give a_0 a factor $\log t$ for each of them, but their second derivatives in M are then of size $\log t$ as well, and the factor $(1 + M)^{n/2}$ can cancel the gain.

All of this comes from measuring derivatives at unit scale in η, while the integral lives at scale $\lambda^{-1/2}$, and at scale $(\lambda \log t)^{-1/2}$ in borderline directions. At that scale Lemma 2 is exactly what is needed: a further derivative in a bad direction m costs $\lambda^{-1/2}/|\eta_m|$, which is at most 1 once $|\eta_m| \geq \lambda^{-1/2}$. Only thin strips around the hyperplanes $\eta_m = 0$ remain. A strip has width about $\lambda^{-1/2}$, the same factor oscillation would give, so it can be integrated trivially. We now carry this out.

5.3 The zoom scale

Early and late switch. If there is a bad direction, we say that the switch is *early* if $|\xi_k| \geq 4t^{-\theta}$, and *late* otherwise. By (3) and the switch time of Section 3, an early switch means that the weight $s^{\ell_k}|\xi_k|$ overtakes every bad weight before the time $s = t^{1/2}$; this is made precise in part (a) of the proof below. If there is no bad direction, there is no borderline direction either, and the distinction plays no role.

Active candidates and scales. The directions used for stationary phase are

$$\mathcal{A} = \mathcal{O} \cup \mathcal{Z} \ \text{ (early switch)}, \qquad \mathcal{A} = \mathcal{O} \ \text{ (late switch)}.$$

Flat directions, and borderline directions after a late switch, are never used for stationary phase. The curvature scale of direction j is

$$L_j = \log t \ \text{ if } j \in \mathcal{Z} \text{ and the switch is early}, \qquad L_j = 1 \ \text{ otherwise}.$$

We take $t_0 \geq e$, so that $L_j \geq 1$.

The zoom. Put $\eta_j = (\lambda L_j)^{-1/2}\omega_j$ for every $j \neq k$. Then

$$J = \lambda^{-n/2} \prod_{j\neq k} L_j^{-1/2} \int_{\mathbb{R}^n} e^{i\tilde{\Phi}(\omega)}\, \tilde{b}(\omega)\, d\omega, \qquad \tilde{\Phi}(\omega) = \lambda\, \Phi(\eta), \quad \tilde{b}(\omega) = b(\xi_k, \eta),$$

and by the chain rule

$$\partial_\omega^\alpha \tilde{\Phi} = \lambda^{1-|\alpha|/2} \prod_{m\neq k} L_m^{-\alpha_m/2}\, \partial_\eta^\alpha \Phi.$$

For $|\alpha| = 2$ this reads $\partial_{\omega_i}\partial_{\omega_j}\tilde{\Phi} = (L_iL_j)^{-1/2}\partial_{\eta_i}\partial_{\eta_j}\Phi$: the Hessian is normalized by the curvature scales. The decay $\lambda^{-n/2}$ and the gain $\prod_j L_j^{-1/2}$ now sit in front, and the ω integral only has to be bounded. This is why borderline directions are zoomed by $(\lambda \log t)^{1/2}$ rather than $\lambda^{1/2}$: the $\log t$ that Lemma 1 puts into a_0 moves into the Jacobian, where nothing can cancel it.

Strips. Let $\mathcal{B}$ be the set of bad directions in $\mathcal{A}$: the bad oscillatory ones, and also the borderline ones after an early switch. Take $\vartheta \in C_0^\infty(\mathbb{R})$ with $\vartheta = 1$ on $[-2, 2]$ and $\vartheta = 0$ outside $[-3, 3]$. For each $m \in \mathcal{B}$ write $1 = \vartheta(\omega_m) + (1 - \vartheta(\omega_m))$ and expand. This splits $J = \sum_S J_S$ over the subsets $S \subset \mathcal{B}$ of directions placed in strips, with

$$J_S = \lambda^{-n/2} \prod_{j\neq k} L_j^{-1/2} \int_{\mathbb{R}^n} e^{i\tilde{\Phi}}\, \tilde{b}_S\, d\omega, \qquad \tilde{b}_S = \tilde{b} \prod_{m\in S} \vartheta(\omega_m) \prod_{m\in\mathcal{B}\setminus S} \big(1 - \vartheta(\omega_m)\big).$$

In J_S the *active* directions are $A = \mathcal{A} \setminus S$. All other directions, those in S and those outside $\mathcal{A}$, are *frozen*.

5.4 Checking the conditions at the zoom scale

Lemma 4 (Conditions at the zoom scale)**.** *Let $t \geq t_0$ and $S \subset \mathcal{B}$, and fix the frozen variables. View $\tilde{\Phi}$ and $\tilde{b}_S$ as functions of $\omega_A \in \mathbb{R}^{|A|}$, and let $\tilde{K}$, $\tilde{K}_1$, a_0, $\tilde{M}$, $\tilde{N}$ be the quantities of Theorem 3 for them. Then $\nabla_{\omega_A}\tilde{\Phi}$ is injective, and*

$$a_0 \geq c, \qquad \tilde{M}_{|A|+2} \leq C, \qquad \tilde{N}_{|A|+1} \leq C,$$

with c, C depending only on d and the ℓ_j.

Proof. We first locate $\tilde{K}_1$ in the η variables. On $\tilde{K}$ the amplitude does not vanish, so $|\eta| \le 2\sqrt{d}$ and (3) holds. Passing to $\tilde{K}_1$ moves each active η_j by at most $(\lambda L_j)^{-1/2} \le 1$. For an active bad m this move is

$$(\lambda L_m)^{-1/2} = L_m^{-1/2}\, t^{-\kappa_m/2} |\xi_k|^{1/2} \cdot \frac{t^{\ell_m - \ell_k}}{|\xi_k|} \le \sqrt{2}\, \frac{t^{\ell_m - \ell_k}}{|\xi_k|},$$

because $\kappa_m \ge 0$ and $|\xi_k| \le 2$. Hence on $\tilde{K}_1$

$$|\eta| \le 3\sqrt{d}, \qquad |\eta_m| \le \frac{4\, t^{\ell_m - \ell_k}}{|\xi_k|} \quad (m \text{ bad}). \tag{7}$$

(a) The determinant a_0. By (7) and Lemma 1 with $R = 3\sqrt{d}$, $-\langle \nabla_\eta^2 \Phi\, q, q\rangle \ge c \sum_j \Gamma_j(\sigma_0) q_j^2$ on $\tilde{K}_1$. We claim $\Gamma_j(\sigma_0) \ge c\, L_j$ for $j \in A$. If $L_j = 1$ this is Lemma 1(i). If $L_j = \log t$, then j is borderline and the switch is early, and (7) gives for every bad m

$$|\eta_m|^{1/(\ell_k - \ell_m)} \le \Big(\frac{4}{|\xi_k|}\Big)^{1/(\ell_k - \ell_m)} t^{-1} \le t^{-1/2},$$

because $|\xi_k| \ge 4t^{-\theta} \ge 4t^{-(\ell_k - \ell_m)/2}$. So $\sigma_*(\eta) \le t^{-1/2}$, and Lemma 1(ii) gives $\Gamma_j(\sigma_0) \ge \frac{1}{2} \log t$. Now let $r \in \mathbb{R}^{|A|}$ and put $q_j = L_j^{-1/2} r_j$ for $j \in A$ and $q_j = 0$ otherwise. By the formula for the Hessian in ω,

$$-\langle \nabla^2_{\omega_A} \tilde{\Phi}\, r, r\rangle = -\langle \nabla^2_\eta \Phi\, q, q\rangle \ge c \sum_{j \in A} \frac{\Gamma_j(\sigma_0)}{L_j}\, r_j^2 \ge c\, |r|^2.$$

So every eigenvalue of $-\nabla^2_{\omega_A} \tilde{\Phi}$ is at least c, and $a_0 \ge c^{|A|}$. In the η variables this says $|\det \nabla^2_{\eta_A} \Phi| \ge c\, (\log t)^{|A \cap \mathcal{Z}|}$: the log in the determinant is exactly the gain $\prod_j L_j^{-1/2}$ in front of J_S.

(b) Injectivity. $\tilde{\Phi}$ is strictly concave in ω_A on all of $\mathbb{R}^{|A|}$, by Section 3 and the linear change of variables. So for $\omega \ne \omega'$,

$$\big\langle \nabla\tilde{\Phi}(\omega) - \nabla\tilde{\Phi}(\omega'), \omega - \omega' \big\rangle = \int_0^1 \big\langle \nabla^2 \tilde{\Phi}\big(\omega' + s(\omega - \omega')\big)(\omega - \omega'), \omega - \omega' \big\rangle\, ds < 0,$$

and $\nabla_{\omega_A} \tilde{\Phi}$ is injective.

(c) The phase derivatives $\tilde{M}$. Let $2 \le |\alpha| \le |A| + 2$ with α supported in $A \subset \mathcal{O} \cup \mathcal{Z}$, and choose i, j in the support of α. By (2), the chain rule and Lemma 2,

$$|\partial_\omega^\alpha \tilde{\Phi}| \le C \Big(\frac{\Gamma_i(1/t)\, \Gamma_j(1/t)}{L_i L_j}\Big)^{1/2} \prod_{m \ne k} \big((\lambda L_m)^{1/2} v_m\big)^{-\alpha'_m}.$$

The first factor is bounded by (5), since $L_m = 1$ for active oscillatory m and $L_m = \log t$ for active borderline m. In the product, a good m has $(\lambda L_m)^{1/2} v_m \ge \lambda^{1/2} \ge 1$. An active bad m lies in $\mathcal{B} \setminus S$ and has $(\lambda L_m)^{1/2} v_m = |\omega_m|$. Since $1 - \vartheta(\omega_m)$ vanishes for $|\omega_m| \le 2$ and $\tilde{K}_1$ extends the support by sup distance 1, we have $|\omega_m| \ge 1$ on $\tilde{K}_1$. So every factor is at most 1, and $\tilde{M}_{|A|+2} \le C$.

(d) The amplitude derivatives $\tilde{N}$. One ω_j derivative, $j \in A$, is $(\lambda L_j)^{-1/2} \partial_{\eta_j}$. Falling on $(1 + |\eta|^2)^{-1/2}$ or on ρ_k, it costs at most $C(\lambda L_j)^{-1/2} \le C$. Falling on $\tilde{\chi}$, it produces the factor $t^{\ell_k - \ell_j} \xi_k$, so it costs

$$(\lambda L_j)^{-1/2}\, t^{\ell_k - \ell_j} |\xi_k| = L_j^{-1/2}\, t^{-\kappa_j/2}\, |\xi_k|^{1/2} \le \sqrt{2},$$

because $\kappa_j \ge 0$, $L_j \ge 1$ and $|\xi_k| \le 2$. This is where $\kappa_j \ge 0$ is used. Derivatives of the strip cutoffs are $O(1)$. So $\tilde{N}_{|A|+1} \le C$. □

The check in one table:

	unit scale in η	zoom scale in ω
a_0	$\geq c\,(\log t)^{\lvert A\cap\mathcal{Z}\rvert}$, offset by M	$\geq c$, gain moved to the Jacobian
M	grows with t in bad directions	$\leq C$ off the strips
N	up to $t^{d(\ell_k-\ell_{\min})}$	$\leq C$ in $\mathcal{O}\cup\mathcal{Z}$

6 Proof of the main estimate

6.1 Statement

Recall from Section 2.4 that $b(\xi_k,\cdot)$ is smooth and supported in $|\eta|\leq 2\sqrt{d}$, and that $|\xi_k|\leq 2$ and $\lambda\geq c\,t^{1+\ell_{\min}}$ whenever $b(\xi_k,\cdot)$ is not identically zero.

Theorem 5 (Main estimate)**.** *There are t_0 and C, depending only on d and $\ell_1,\dots,\ell_d$, such that for $t\geq t_0$, all $x\in\mathbb{R}^d$ and all $\xi_k\neq 0$,*

$$\Big|\int_{\mathbb{R}^n} e^{i\lambda\Phi(\eta)}\,b(\xi_k,\eta)\,d\eta\Big|\leq C\,\lambda^{-n/2}\prod_{j\in\mathcal{F}}t^{\kappa_j/2}\,|\xi_k|^{-1/2}\prod_{j\in\mathcal{Z}}E(\xi_k),\qquad \lambda=t^{\ell_k+1}|\xi_k|,$$

where $E(\xi_k)=(\log t)^{-1/2}$ if $|\xi_k|\geq 4t^{-\theta}$ and $E(\xi_k)=|\xi_k|^{-1/2}$ otherwise.

Each factor in the product over $\mathcal{F}$ is at most $|\xi_k|^{-1/2}$, since $\kappa_j<0$ there. When every direction is oscillatory, the estimate is the full stationary phase rate $\lambda^{-n/2}$. Each borderline direction gives an extra factor $(\log t)^{-1/2}$ after an early switch. After a late switch it is treated like a flat direction with $\kappa_j=0$.

Remark. The factors $|\xi_k|^{-1/2}$ have to be handled in the outer ξ_k integral of I_k. So does the late-switch region $|\xi_k|<4t^{-\theta}$, which has length at most $8t^{-\theta}$.

6.2 Proof

Write J for the integral in Theorem 5, with t, x, ξ_k fixed. If $b(\xi_k,\cdot)\equiv 0$ there is nothing to prove, so we assume $|\xi_k|\leq 2$ and $\lambda\geq c\,t^{1+\ell_{\min}}$, and we take $t_0\geq e$ so large that $\lambda\geq 1$.

Step 1: zoom and strips. By Section 5.3, $J=\sum_S J_S$ over the $2^{|\mathcal{B}|}$ subsets $S\subset\mathcal{B}$, with the factor $\lambda^{-n/2}\prod_j L_j^{-1/2}$ in front of each J_S.

Step 2: stationary phase in the active variables. Fix S and the frozen variables. By Lemma 4, Theorem 3 applies to the ω_A integral with $\lambda=1$ and gives

$$\Big|\int_{\mathbb{R}^{|A|}} e^{i\tilde{\Phi}}\,\tilde{b}_S\,d\omega_A\Big|\leq C,$$

uniformly in x and in the frozen variables.

Step 3: trivial integration in the frozen variables.

- $m\in S$. The cutoff $\vartheta(\omega_m)$ is supported in $|\omega_m|\leq 3$, so this costs a factor 6.

- $j \notin \mathcal{A}$, a flat direction or a borderline one after a late switch. Then $L_j = 1$, and (3) gives $|\omega_j| \le W_j$ with

$$W_j = \lambda^{1/2} \cdot \frac{2t^{\ell_j - \ell_k}}{|\xi_k|} = 2\, t^{\kappa_j/2}\, |\xi_k|^{-1/2} \le 2\, |\xi_k|^{-1/2},$$

so this costs a factor $2W_j$. Here $\kappa_j \le 0$ is used: the support in ω_j is no wider than a constant times $|\xi_k|^{-1/2}$.

Step 4: conclusion. Each piece satisfies

$$|J_S| \le C\, \lambda^{-n/2} \prod_{j \ne k} L_j^{-1/2}\, 6^{|S|} \prod_{j \notin \mathcal{A}} 2W_j.$$

After an early switch, $\prod_j L_j^{-1/2} = (\log t)^{-|\mathcal{Z}|/2}$ and the directions outside $\mathcal{A}$ are the flat ones. After a late switch, all $L_j = 1$ and the directions outside $\mathcal{A}$ are the flat and the borderline ones, with $t^{\kappa_j/2} = 1$ for the latter. Summing over S proves Theorem 5. The bound is uniform in x because x enters Φ only through its linear part, which none of a_0, $\tilde{M}$, $\tilde{N}$ sees.

Summary. After zooming to $\omega_j = (\lambda L_j)^{1/2}\eta_j$, the condition $\kappa_j \ge 0$ keeps the amplitude smooth at scale 1, and Lemma 1 keeps the Hessian uniformly nondegenerate. Oscillatory directions gain $\lambda^{-1/2}$, and borderline directions gain $(\lambda \log t)^{-1/2}$ after an early switch. Flat directions are thinner than the stationary phase scale and are integrated trivially, and so are the strips where the weighted bounds degenerate.